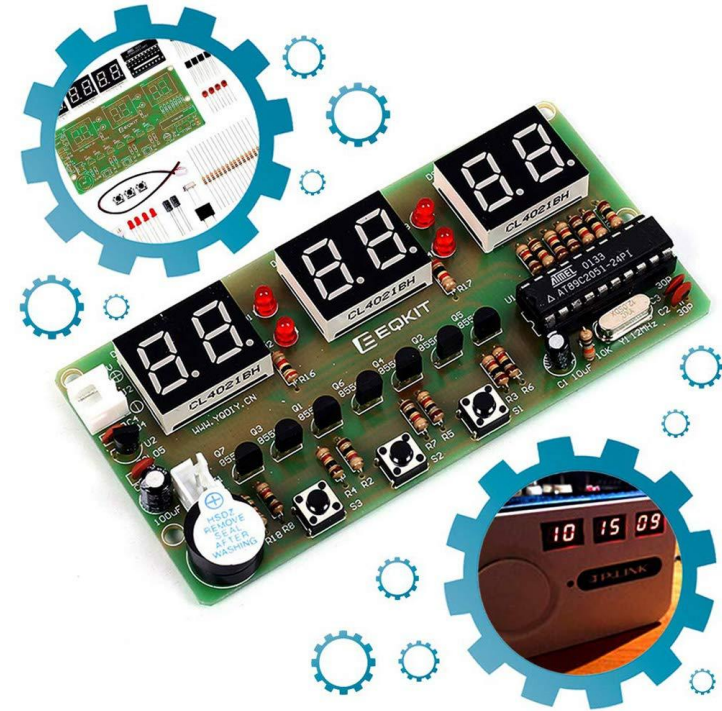




Dr. Hiran Ekanayake

DIGITAL ELECTRONIC FUNDAMENTALS – PART 2

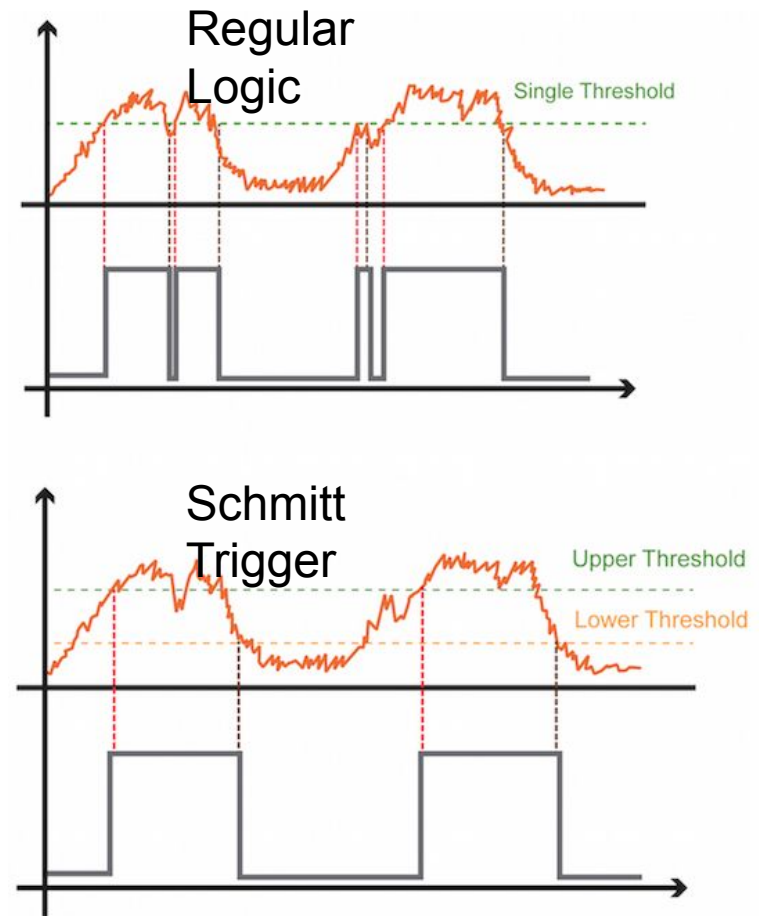
Lesson Outline

- AND, OR, and NOT logic functions
- Digital logic ICs: logic levels, IC families, data sheets, etc.
- Schmitt Trigger
- NAND, NOR, XOR, and XNOR logic functions
- Digital Buffers
- Arduino-based Logic Function Detector

SCHMITT TRIGGER

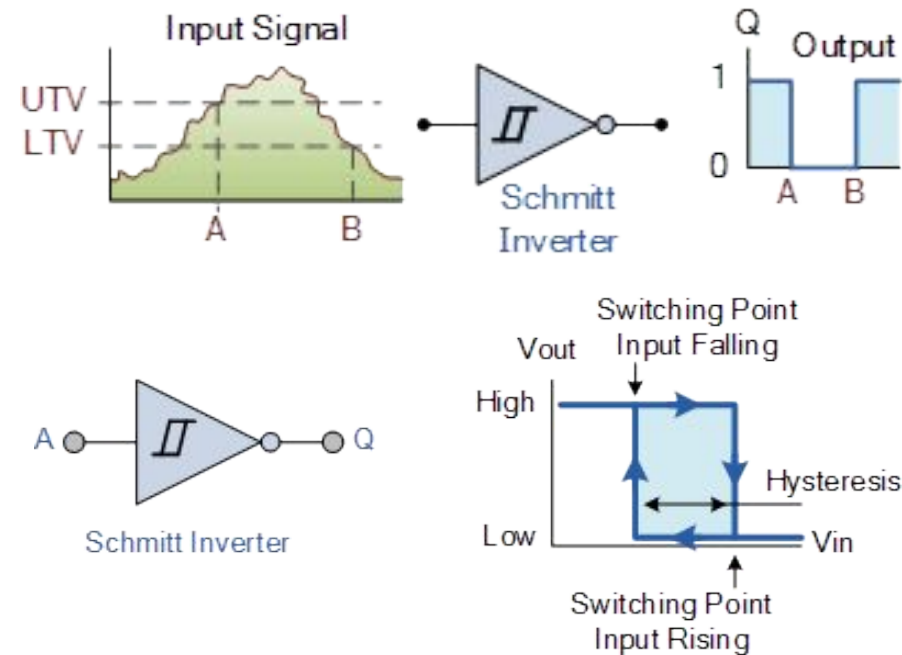
Schmitt Trigger

- Briefly explain how a Schmitt inverter minimizes the effects of noise in an input signal.
 - A Schmitt Inverter is designed to operate or switch state when its input signal goes above an “Upper Threshold Voltage” or UTV limit in which case the output changes and goes “LOW” and will remain in that state until the input signal falls below the “Lower Threshold Voltage” or LTV level in which case the output signal goes “HIGH”.



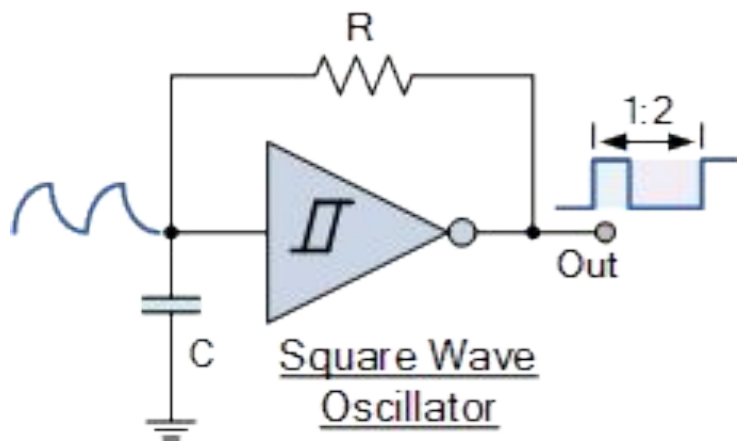
Schmitt Trigger

- Briefly explain how a Schmitt inverter minimizes the effects of noise in an input signal.
 - A Schmitt Inverter is designed to operate or switch state when its input signal goes above an “Upper Threshold Voltage” or UTV limit in which case the output changes and goes “LOW” and will remain in that state until the input signal falls below the “Lower Threshold Voltage” or LTV level in which case the output signal goes “HIGH”.

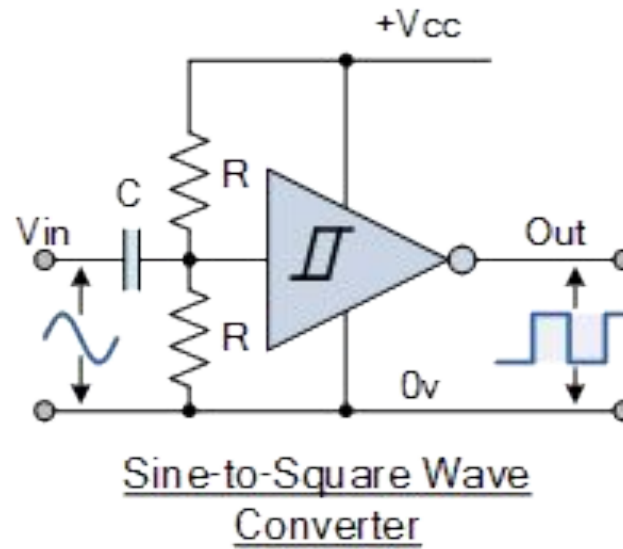


It provides a much cleaner and faster “ON/OFF” switching output signal

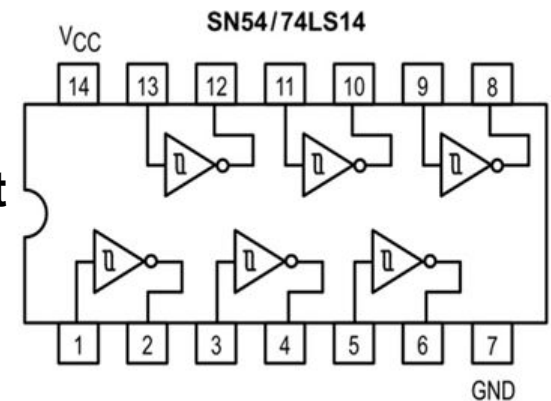
Applications of Schmitt Inverter

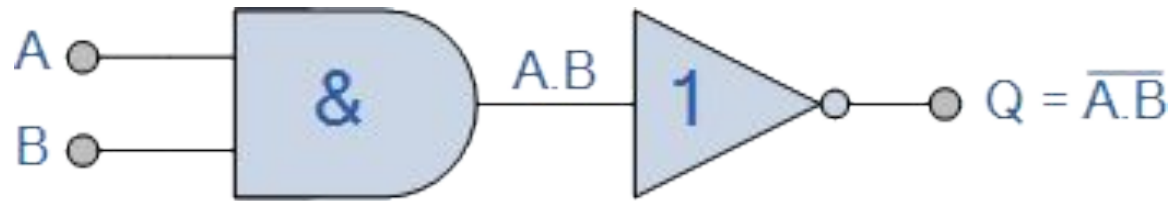


$$f = 680/RC.$$



Hex Schmitt Inverter





2-input "AND" gate plus a "NOT" gate

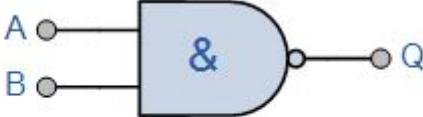


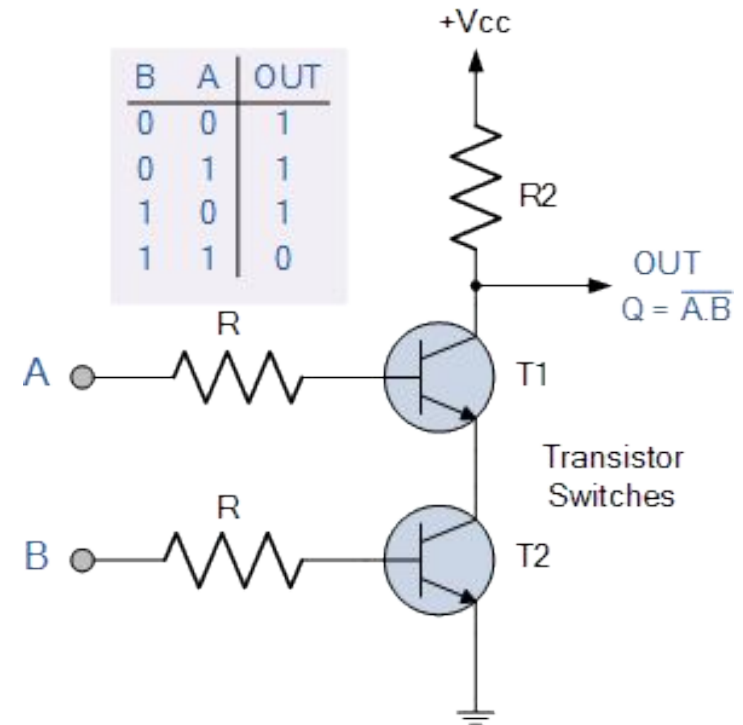
LOGIC NAND

“The Logic NAND Function output is normally true and only goes to false when all of its inputs are true”

“Compliment/Inverse of AND”

LOGIC NAND

Symbol	Truth Table		
 2-input NAND Gate	B	A	Q
	0	0	1
	0	1	1
	1	0	1
	1	1	0
Boolean Expression $Q = \overline{A \cdot B}$	Read as A AND B gives NOT Q		



Sheffer Stroke Function

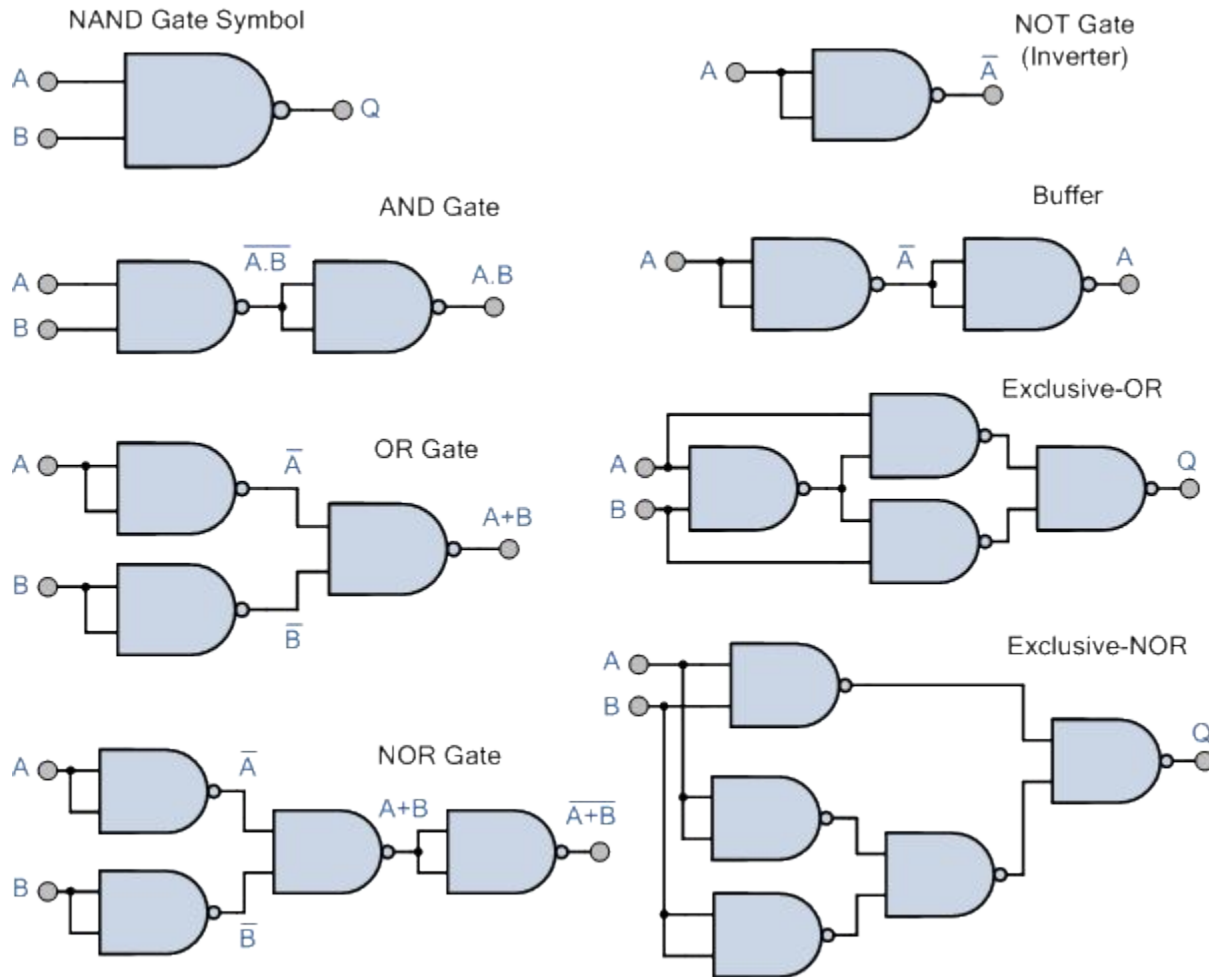
- Logic NAND function is sometimes known as the Sheffer Stroke Function, denoted by $A|B$ or $A \uparrow B$
- Why? NAND is **functionally complete**, i.e., it can be used to produce any other type of logic gate function; thus, it is considered as a “universal” gate

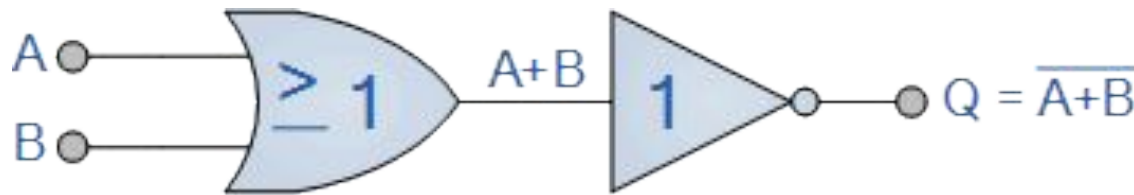
$$\text{NOT } A = A \uparrow A$$

$$A \text{ AND } B = (A \uparrow B) \uparrow (A \uparrow B)$$

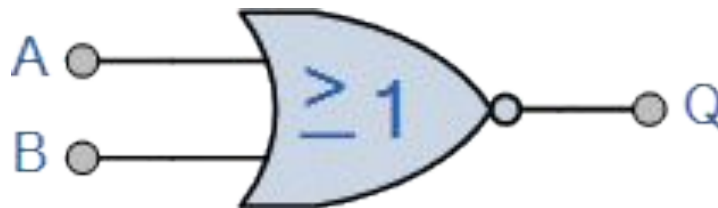
$$A \text{ OR } B = (A \uparrow A) \uparrow (B \uparrow B)$$

Other Logic Functions Using NAND Gate





2-input "OR" gate plus a "NOT" gate

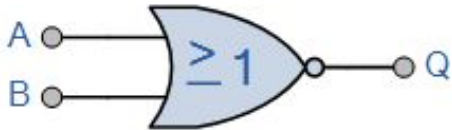


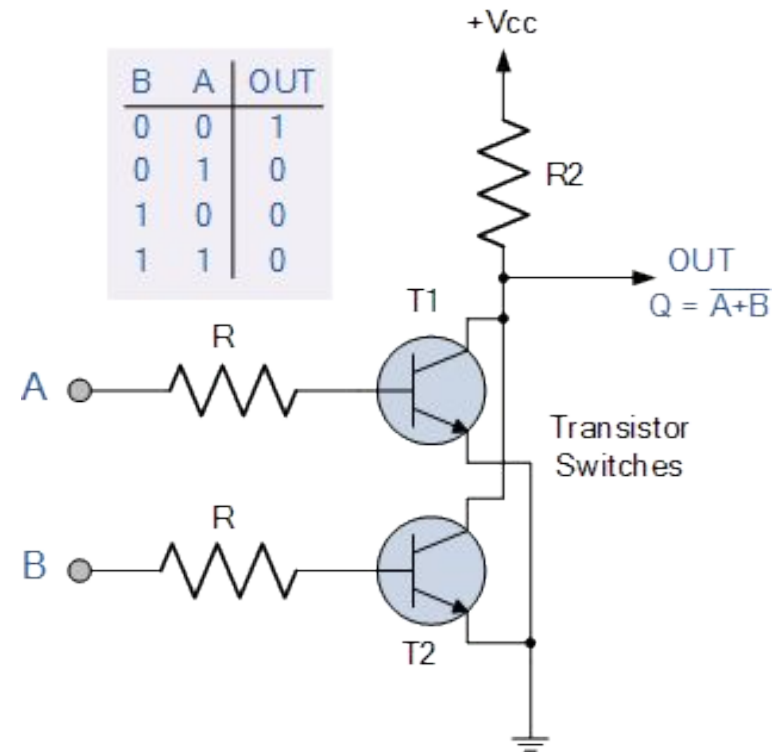
LOGIC NOR

“The Logic NOR Function output is normally true and only goes to false when any of its inputs are true”

“Compliment/Inverse of OR”

LOGIC NOR

Symbol	Truth Table		
 2-input NOR Gate	B	A	Q
	0	0	1
	0	1	0
	1	0	0
	1	1	0
Boolean Expression $Q = \overline{A+B}$	Read as A OR B gives NOT Q		



Pierce Function

- Logic NOR function is sometimes known as the Pierce Function, denoted by $A \downarrow B$
- Why? NOR is **functionally complete**, i.e., it can be used to produce any other type of logic gate function; thus, it is considered a “universal” gate

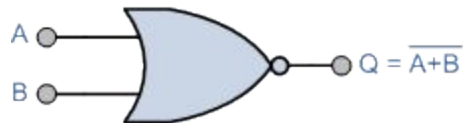
$$\text{NOT } A = A \downarrow A$$

$$A \text{ OR } B = (A \downarrow B) \downarrow (A \downarrow B)$$

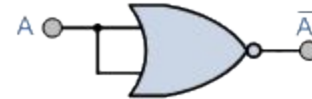
$$A \text{ AND } B = (A \downarrow A) \downarrow (B \downarrow B)$$

Other Logic Using NOR Gate

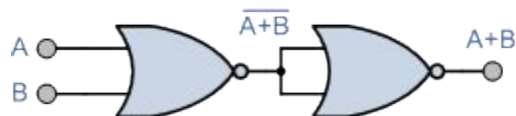
NOR Gate Symbol



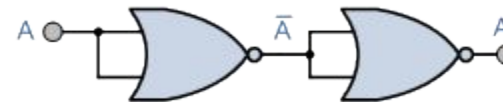
NOT Gate
(Inverter)



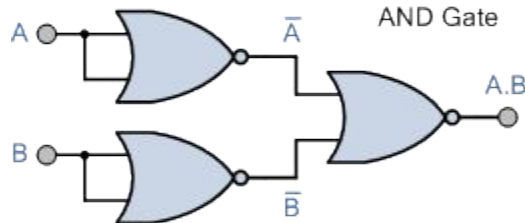
OR Gate



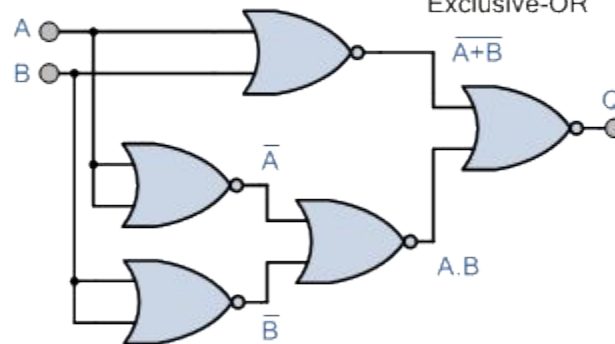
Buffer



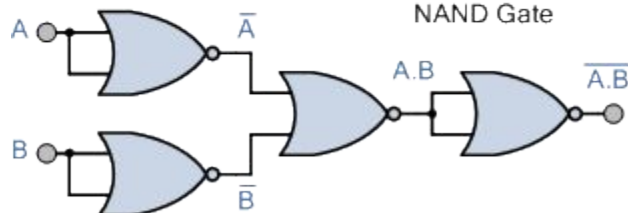
AND Gate



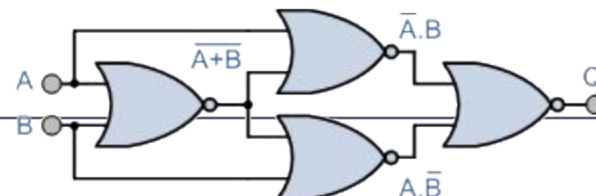
Exclusive-OR

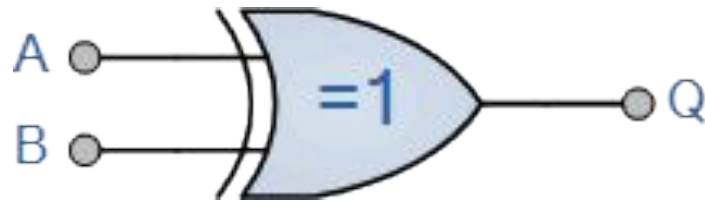


NAND Gate



Exclusive-NOR

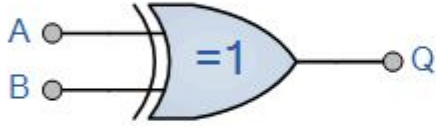




EXCLUSIVE-OR (XOR)

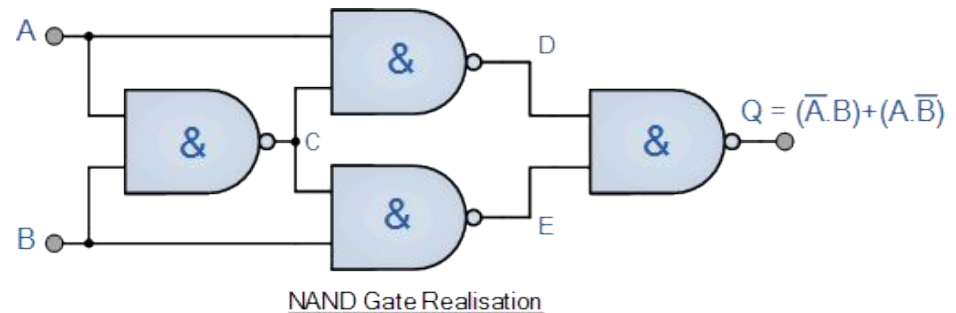
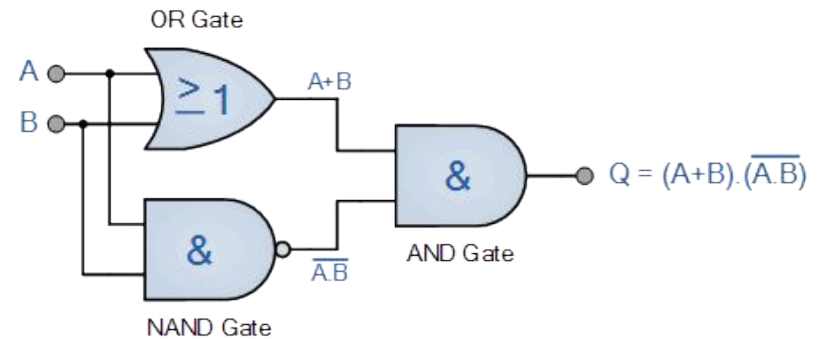
“The Logic XOR Function output is only true when its two input terminals are at different logic levels with respect to each other.”

LOGIC XOR

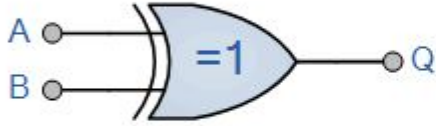
Symbol	Truth Table		
 2-input Ex-OR Gate	B	A	Q
	0	0	0
	0	1	1
	1	0	1
	1	1	0
Boolean Expression $Q = A \oplus B$	A OR B but NOT BOTH gives Q		

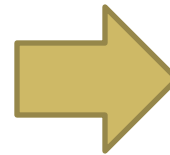
$$Q = (A \oplus B) = \bar{A}.B + A.\bar{B}$$

$$A \oplus B = (A \wedge \neg B) \vee (\neg A \wedge B)$$



More Than 2-Inputs?

Symbol	Truth Table		
 2-input Ex-OR Gate	B	A	Q
	0	0	0
	0	1	1
	1	0	1
	1	1	0
Boolean Expression $Q = A \oplus B$	A OR B but NOT BOTH gives Q		



“A OR B but NOT both”

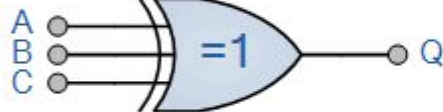
“Odd Function”

“Mod-2-SUM”

$$Q = (A \oplus B) = \bar{A}.B + A.\bar{B}$$

More Than 2-Inputs?

A
B



3-input Ex-OR Gate

Bool

Boolean Expression $Q = A \oplus B \oplus C$

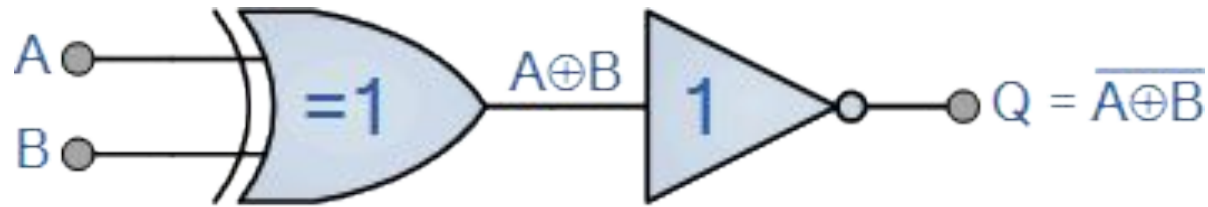
"Any ODD Number of Inputs" gives Q

"A OR B but NOT both"

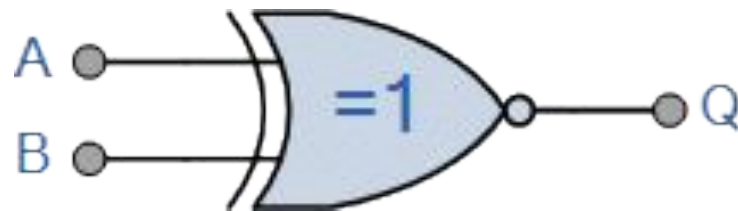
"Odd Function"

"Mod-2-SUM"

$$Q = A\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C + ABC$$



2-input "Ex-OR" gate plus a "NOT" gate

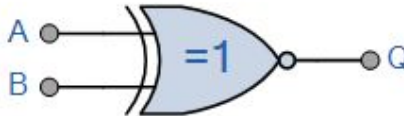


EXCLUSIVE-NOR (XNOR)

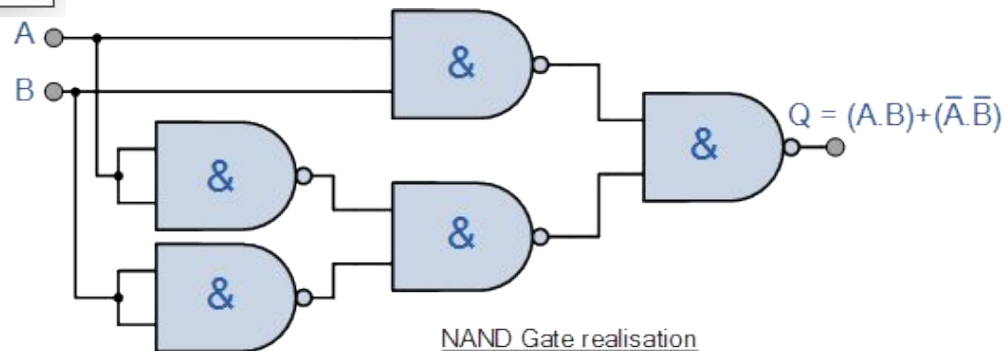
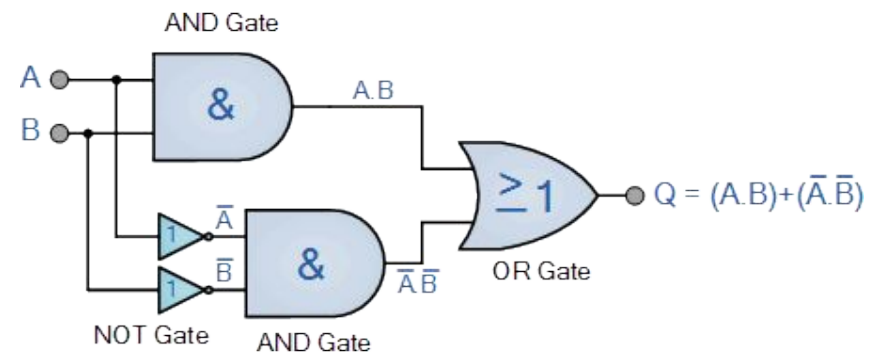
“The Logic XNOR Function output is only true when both of its inputs are at the same/equal logic level”

“Compliment/Inverse of XOR”

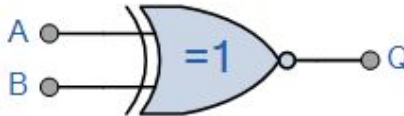
LOGIC XNOR

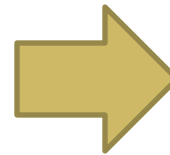
Symbol	Truth Table		
 2-input Ex-NOR Gate	B	A	Q
	0	0	1
	0	1	0
	1	0	0
	1	1	1
Boolean Expression $Q = \overline{A \oplus B}$	Read if A AND B the SAME gives Q		

$$Q = \overline{AB} + AB$$



More Than 2-Inputs?

Symbol	Truth Table		
 2-input Ex-NOR Gate	B	A	Q
	0	0	1
	0	1	0
	1	0	0
	1	1	1
Boolean Expression $Q = \overline{A} \oplus B$	Read if A AND B the SAME gives Q		



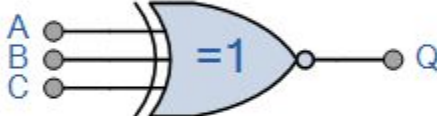
“when both A AND B are the SAME”

“Even Function”

“Mod-2-SUM”

$$Q = \overline{A}B + AB$$

More Than 2-Inputs?

	Symbol	Truth Table			
		C	B	A	Q
A	 3-input Ex-NOR Gate	0	0	0	1
B		0	0	1	0
		0	1	0	0
		0	1	1	1
		1	0	0	0
		1	0	1	1
		1	1	0	1
		1	1	1	0
Boolean Expression		$Q = \overline{A \oplus B \oplus C}$			
		Read as "any EVEN number of Inputs" gives Q			

“when both A AND B are the SAME”

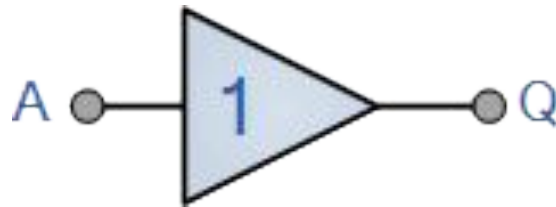
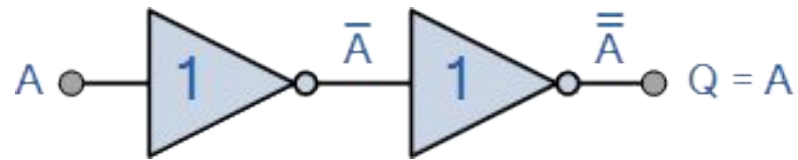
“Even Function”

“Mod-2-SUM”

$$Q = \overline{ABC} + A\overline{B}\overline{C} + A\overline{B}C + A\overline{B}C$$

Digital Logic Gate ICs

	TTL	CMOS
AND	74LS08 Quad 2-input 74LS11 Triple 3-input 74LS21 Dual 4-input	CD4081 Quad 2-input CD4073 Triple 3-input CD4082 Dual 4-input
OR	74LS32 Quad 2-input	CD4071 Quad 2-input CD4075 Triple 3-input CD4072 Dual 4-input
NOT	74LS04 Hex Inverting NOT Gate 74LS14 Hex Schmitt Inverting NOT Gate 74LS1004 Hex Inverting Drivers	CD4009 Hex Inverting NOT Gate CD4069 Hex Inverting NOT Gate
NAND	74LS00 Quad 2-input 74LS10 Triple 3-input 74LS20 Dual 4-input 74LS30 Single 8-input	CD4011 Quad 2-input CD4023 Triple 3-input CD4012 Dual 4-input
NOR	74LS02 Quad 2-input 74LS27 Triple 3-input 74LS260 Dual 4-input	CD4001 Quad 2-input CD4025 Triple 3-input CD4002 Dual 4-input
XOR	74LS86 Quad 2-input	CD4030 Quad 2-input
XNOR	74LS266 Quad 2-input	CD4077 Quad 2-input

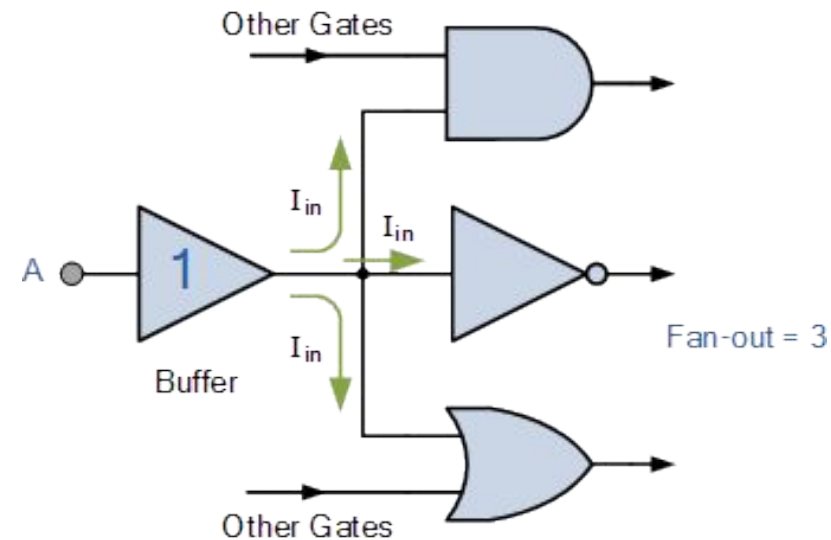


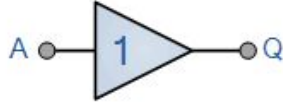
DIGITAL BUFFER

“Q is true, only when A is true”

Digital Buffer

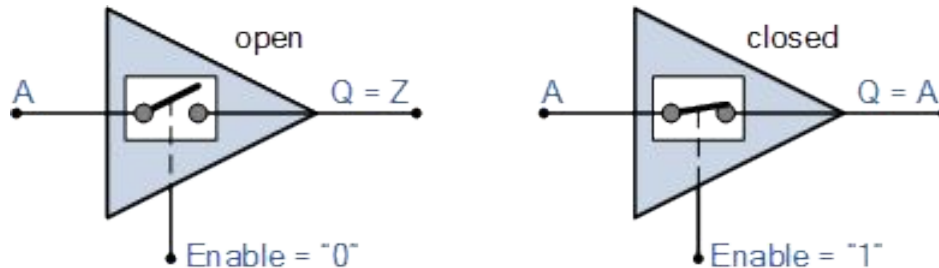
- What is a digital buffer?
What is the purpose of it?
 - It can provide current amplification in a digital circuit to drive output loads, such as relays, solenoids and lamps (called “**fan-out**”)
 - It can be used to isolate other gates or circuit stages from each other preventing the **impedance** of one circuit from affecting the impedance of another



Symbol	Truth Table	
 The Digital Buffer	A	Q
	0	0
	1	1
Boolean Expression $Q = A$		Read as: A gives Q

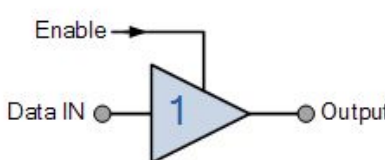
Tri-State Buffer

- A digital buffer with a control (enable) switch
 - Input: 0 or 1
 - Output: when $EN=1$ $\square$ 0 or 1 depending on the input, and when $EN=0$ $\square$ “Hi-Z” (output off, no logic, high-impedance disconnect)



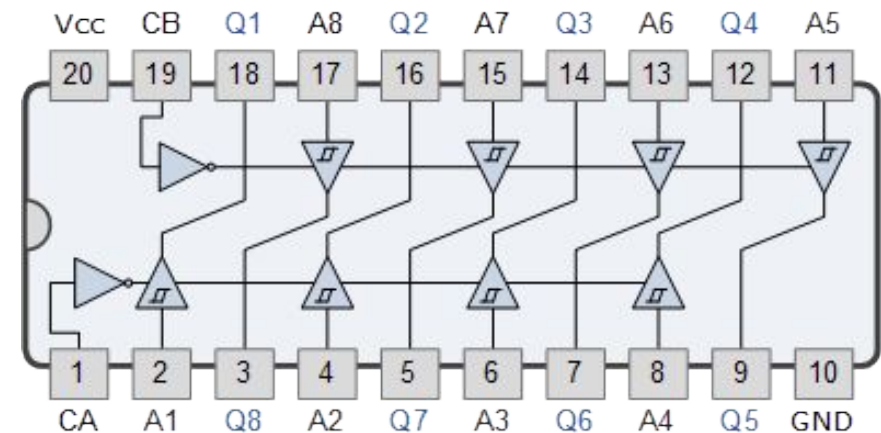
- Why “Hi-Z” is important?
 - There is a limit to the number of inputs and outputs that can be connected together in digital elements, e.g., a buffer has a fan-out limit of about 20

Types of Tri-State Buffers

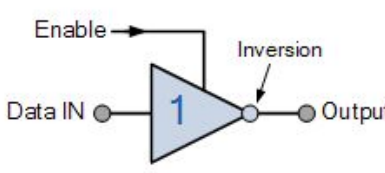
Symbol	Truth Table		
 Tri-state Buffer	Enable	IN	OUT
	0	0	Hi-Z
	0	1	Hi-Z
	1	0	0
	1	1	1
Read as Output = Input if Enable is equal to "1"			

Active-high Tri-state Buffer

74LS244 Octal Tri-state Buffer



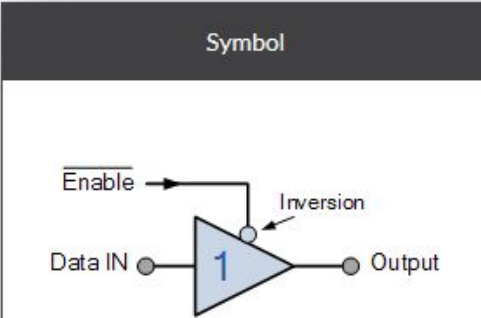
Types of Tri-State Buffers

Symbol	Truth Table															
 Inverting Tri-state Buffer	<table><tr><th>Enable</th><th>IN</th><th>OUT</th></tr><tr><td>0</td><td>0</td><td>Hi-Z</td></tr><tr><td>0</td><td>1</td><td>Hi-Z</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	Enable	IN	OUT	0	0	Hi-Z	0	1	Hi-Z	1	0	1	1	1	0
Enable	IN	OUT														
0	0	Hi-Z														
0	1	Hi-Z														
1	0	1														
1	1	0														
Read as Output = Inverted Input if Enable equals "1"																

Active-high Tri-state Buffer

Active-high inverting tri-state buffer

Types of Tri-State Buffers

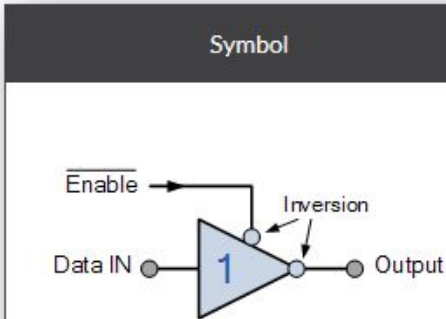
Symbol		Truth Table		
 Tri-state Buffer		Enable	IN	OUT
		0	0	0
		0	1	1
		1	0	Hi-Z
		1	1	Hi-Z
Read as Output = Input if Enable is NOT equal to "1"				

Active-high Tri-state Buffer

Active-high inverting tri-state buffer

Active-low Tri-state Buffer

Types of Tri-State Buffers

Symbol		Truth Table		Active-high Tri-state Buffer Active-high inverting tri-state buffer Active-low Tri-state Buffer Active-low Inverting Tri-state Buffer													
Symbol		Truth Table															
Symbol		Truth Table															
Symbol		Truth Table															
 Inverting Tri-state Buffer		<table><tr><th>Enable</th><th>IN</th><th>OUT</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>Hi-Z</td></tr><tr><td>1</td><td>1</td><td>Hi-Z</td></tr></table>	Enable	IN	OUT	0	0	1	0	1	0	1	0	Hi-Z	1	1	Hi-Z
Enable	IN	OUT															
0	0	1															
0	1	0															
1	0	Hi-Z															
1	1	Hi-Z															
Read as Output = Inverted Input if Enable is NOT equal to "1"																	

Application: Data Bus Control

